How quickly do fixed compute thresholds lose track of capable AI models?

Devika Chandiran

Independent researcher, Liverpool, UK | devikachandiran135@gmail.com | github.com/devikachandiran

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Summary. In this study, I wanted to find out whether fixed compute thresholds can still identify highly capable AI models as AI technology continues to improve. To explore this, I used Epoch AI's dataset, accounted for improvements in algorithmic efficiency, and ran 2,000 simulations to see how uncertainty might affect the results. What surprised me most was that the 10^{25} FLOP threshold caught only about 15% of capable models in 2025. Based on what I found, I think these thresholds should be reviewed roughly every six months and used alongside tests that assess what AI models can actually do.

1 Introduction

Can governments continue to rely on fixed computing-power thresholds to identify highly capable AI models when AI algorithms are becoming more efficient?

A compute threshold is a fixed amount of training compute used as a cut-off to decide whether an AI model falls under particular regulatory rules. Governments use it because compute is easy to measure, while capability is not, so it gives developers a clear rule. For example, the European Union's AI Act presumes that a general-purpose model has "high impact capabilities" when its training compute is greater than 10^{25} FLOP (Article 51) [3].

However, as algorithms improve, companies can build highly capable models with less compute. A model could therefore fall below the threshold while still having capabilities that warrant safety scrutiny. Earlier work has argued that compute is an imperfect proxy for capability (Heim & Koessler, 2024 [4]; Hooker, 2024 [5]).

In this note I measure how quickly that problem grows. Drawing on my background in gravitational-wave astronomy, I treat the threshold as a detector and estimate its detection efficiency: the share of capable models it captures each year. I then ask how often a threshold would need updating to keep catching them.

2 Data and method

I downloaded Epoch AI's All AI Models dataset [1] on 25 October 2026 to analyse this. I focused on 822 language and multimodal AI models developed from 2018 onwards that had available estimates of their training compute.

One thing I wanted to account for was how much AI algorithms have improved over time. A model today may achieve the same level of capability as an older model while using much less computing power. To account for this, I adjusted each model's training compute using the findings of Ho et al. (2024) [2], who estimated that the compute needed to achieve the same capability halves roughly every eight months, with a 95% confidence interval of around 5–14 months.

I considered a model to be "capable" if its estimated capability was at least equivalent to a model trained using 10^{25} FLOP in January 2023. I then compared different compute thresholds to see what percentage of these capable models each rule would catch.

Since the compute estimates and rate of algorithmic improvement are not exact, I ran 2,000 Monte Carlo simulations to account for uncertainty. I also allowed for greater uncertainty in models that Epoch AI labelled as "Likely" or "Speculative".

This approach is similar to what I used in my gravitational-wave research, where I examined how effectively detectors could identify signals above a certain detection threshold. Here, I applied the same idea to AI models to understand how well compute thresholds identify capable models and whether their effectiveness changes over time.

3 Results

The results show that although AI models are becoming more powerful, a fixed compute threshold is becoming less effective at identifying capable models.

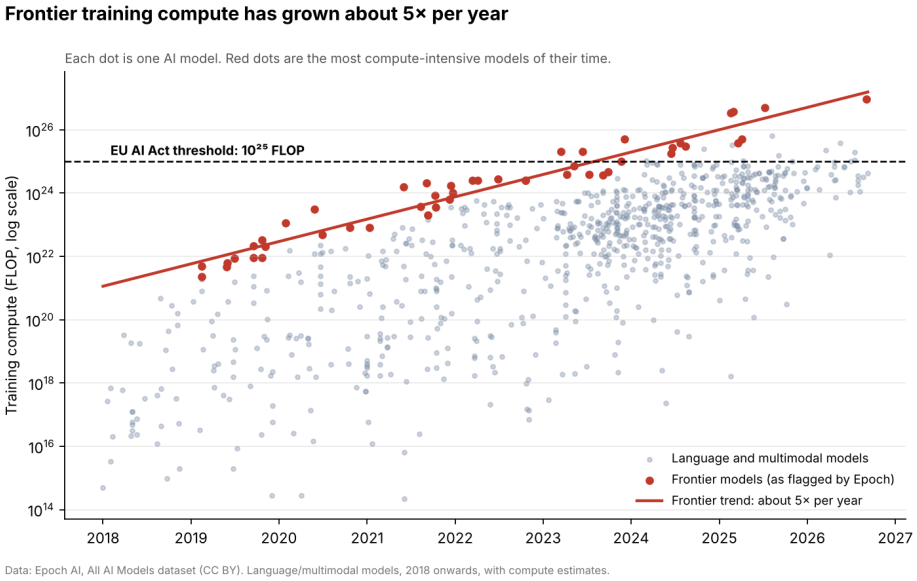


Figure 1: Training compute of 822 language and multimodal models, 2018–2026. The red line is a fit to the models Epoch flags as frontier models. Data: Epoch AI (CC BY).

Figure 1 shows how quickly training compute has increased for frontier AI models. On average, frontier training compute has grown by around five times per year, with a 90% uncertainty range of 4.6–5.6 times. But compute alone does not show how capable these models are. Figure 2 tests that.

Figure 2 shows the main finding of my analysis. In 2023, the 10^{25} FLOP threshold caught approximately 67% of capable models. However, this dropped to around 29% in 2024 and just 15% in 2025, with an uncertainty range of 10–24%. This surprised me because it means that by 2025, around 85% of models meeting my capability definition were falling below the threshold. I also tested what would happen if algorithmic progress was much slower, with compute requirements halving every 24 months instead of eight (Table 1). Even then, coverage in 2025 reached only around 42%, suggesting that the problem does not depend on how quickly algorithms improve.

Table 1: Coverage of the strict 10^{25} FLOP rule in 2025 under different speeds of algorithmic progress (median, with 5–95% range).

Halving time of compute needed	Capable models caught in 2025
5 months	11% (8–14%)
8 months (central estimate)	15% (11–19%)
14 months	26% (19–33%)
24 months	42% (31–54%)

Figure 3 shows what happens when the threshold is lowered. At 10^{24} FLOP, coverage increases to approximately 92%, which is a considerable improvement, and there are almost no false alarms.

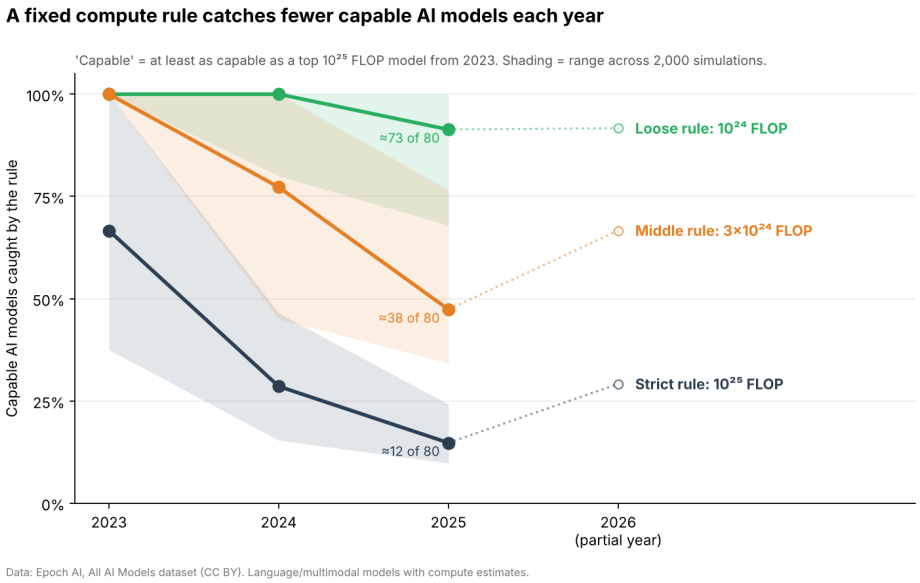


Figure 2: Share of capable models caught by three fixed thresholds. "Capable" means at least as capable as a top 10^{25} FLOP model from January 2023. Shading shows the 5–95% range across 2,000 simulations. 2026 is a partial year. Data: Epoch AI (CC BY).

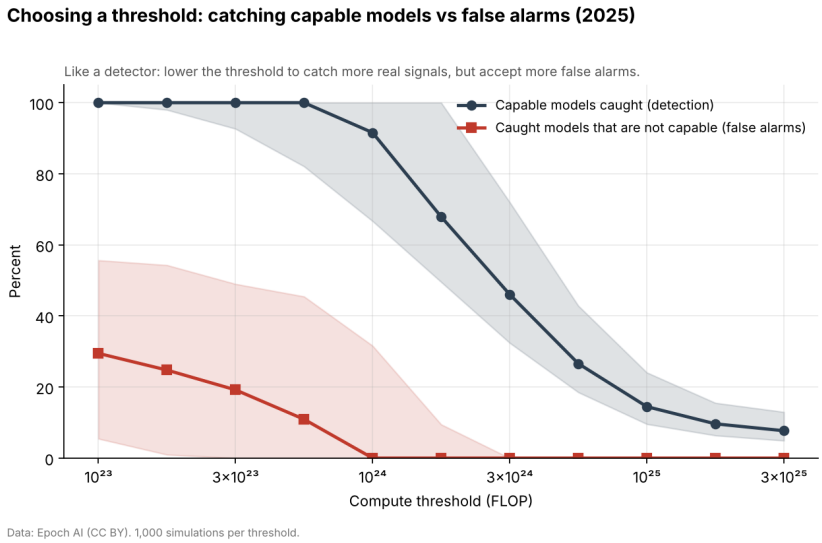


Figure 3: Detection (share of capable models caught) and false alarms (share of caught models that fall below the capability tier) across thresholds, 2025. Shading shows the 5–95% range. Data: Epoch AI (CC BY).

However, lowering the threshold further also means catching more models that do not meet my capability definition. At 10^{23} FLOP, the false-alarm rate rises to around 29%. This highlights the difficulty of finding a threshold that catches most capable models without placing unnecessary regulatory requirements on others.

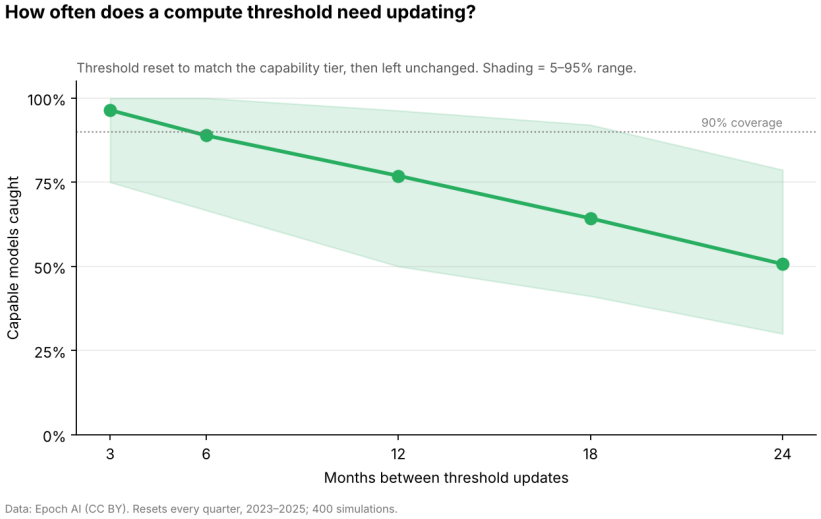


Figure 4: Average share of capable models caught when the threshold is reset to match the capability tier and then left unchanged. Resets every quarter, 2023–2025; 400 simulations. Data: Epoch AI (CC BY).

Figure 4 was particularly interesting to me. Updating the threshold every six months maintains approximately 89% coverage, whereas updating it every 24 months reduces coverage to around 51%. This suggests that how often a threshold is reviewed could be just as important as where it is initially set.

I also compared open-weight and closed models in 2025. The strict threshold caught around 57% of capable closed models but only 6% of capable open-weight models. However, the closed-model sample contained just 14 models, so I would treat this difference cautiously rather than draw a firm conclusion.

Overall, these results suggest that compute thresholds can still be useful, but relying on a fixed threshold for several years could leave many capable AI models outside its coverage.

4 Implications for policy

From this analysis, I think compute thresholds are a good starting point for regulating AI, but they shouldn't be the only thing governments rely on. They can help identify models that might need additional safety checks, but just knowing how much computing power a model uses does not tell us whether it is safe or potentially harmful.

What surprised me most was how quickly these thresholds could become outdated. My results suggest that reviewing and updating them around every six months could make a big difference: in my tests, updating about every six months kept most capable models covered.

I also think governments need to look beyond computing power and pay more attention to what AI models can actually do. For example, testing their capabilities, identifying possible risks and checking how well their safety measures work could help identify models that might otherwise be missed.

At the same time, simply lowering the threshold isn't necessarily the answer. Figure 3 shows that while a lower threshold catches more capable models, it also includes more models that may not

need the same level of regulation. This could create extra costs and work for developers.

Another thing I found important is that being below a compute threshold doesn't mean an AI model is safe. As algorithms improve, companies can develop highly capable models using less computing power than before.

Overall, I think compute alone is becoming a less reliable way of identifying capable AI models. It still has a role in regulation, but it needs to be regularly reviewed and used alongside actual capability and safety assessments.

5 Limitations

There are a few limitations to this analysis that are important to consider. The biggest one is how I defined a "capable" AI model. Instead of using actual capability test scores, I estimated capability based on effective compute. This means the results show how well the thresholds identify models that are expected to be capable, rather than confirming what those models can actually do.

Another limitation is that I assumed algorithmic improvements affect all models in the same way. In reality, some models may benefit more from these improvements than others, depending on their architecture and training methods.

Also, many of the training-compute values in Epoch AI's dataset are estimates rather than officially reported figures, which adds uncertainty to the results.

The 2026 data is still incomplete, so I would be careful about drawing conclusions for that year. Finally, the dataset may not include models that were developed privately or never publicly announced.

Overall, these limitations mean the results should be treated as estimates rather than exact measurements of AI capability or regulatory effectiveness.

6 Next steps

If I had more time to work on this, I would first look at how the compute estimates compare with actual AI benchmark scores. I think this would give me a better idea of how closely computing power is linked to what a model can really do.

I would also like to explore whether looking at potentially harmful capabilities, rather than just general AI performance, would change the results. It would be interesting to compare different types of AI models as well, to see if the same pattern appears across them.

Finally, I would like to repeat the analysis when the complete 2026 data becomes available, especially to see whether the thresholds continue to miss more capable models over time.

References

[1] Epoch AI. *Data on AI Models*. `https://epoch.ai/data/ai-models`, accessed 25 October 2026. Licensed CC BY.

[2] Ho, A., Besiroglu, T., Erdil, E., et al. (2024). *Algorithmic progress in language models*. NeurIPS 2024. arXiv:2403.05812.

[3] European Union (2024). *Regulation (EU) 2024/1689 (Artificial Intelligence Act)*, Article 51.

[4] Heim, L. & Koessler, L. (2024). *Training Compute Thresholds: Features and Functions in AI Regulation*. arXiv:2405.10799.

[5] Hooker, S. (2024). *On the Limitations of Compute Thresholds as a Governance Strategy*. arXiv:2407.05694.